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Sample structured questions

Question 1

Consider the following Binary Search Tree,

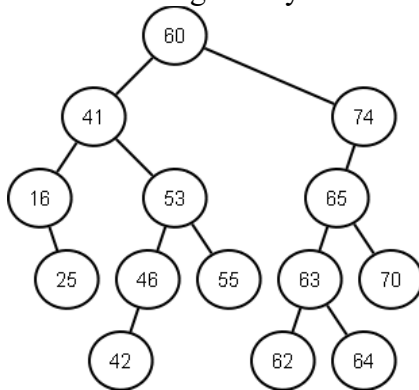


Figure 2: Binary search tree

To search for a specific node in a binary search tree, the following algorithm may be applied.

1. *SearchElement(Tree,Val)*
2. *Step 1: if tree->data= Val or Tree-null*
3. *Return Tree*
4. *Else*
5. *If Val< Tree->Data*
6. *Return (SearchElement(Tree->left,Val)*
7. *Else*
8. *Return (SearchElement(Tree->right,Val)*
9. *[end of if]*
10. *[end of if]*
11. *Step 2: end*

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- (a) Explain the steps of the above algorithm when searching for node 62 in the binary search tree shown above.

(5 Marks)

- b) Explain how the above search is adapted with a minor extension to insert node 61 into the tree

(5 Marks)

- c) Which node(s), if any, can replace a deleted node while preserving binary search tree properties for each of the following deletions from the tree shown in Figure 2.

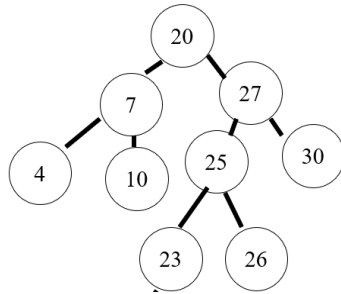
Note: For each deletion, use the original tree shown in Figure 2 above.

(5 Marks)

Node to be deleted	Replacing Node(s), If Any
(i)25	
(ii)48	
(iii)53	
(iv) 60	

- d) Consider the following AVL tree.

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After inserting the new node 22 into the above AVL tree, identify which of the following statement(s) are true:

- i. Imbalance detected at node 25.
- ii. Imbalance detected at node 10
- iii. Imbalance detected at node 26
- iv. Imbalance detected at node 27

(5 Marks)

e) Which of the following statement(s) is/are correct after the above insertion

- i. To establish the AVL properties after inserting 22, a single rotation is recommended.
- ii. To establish the AVL properties after inserting 22, a double rotation is recommended.
- iii. Two opposite single rotations are needed to re-establish the AVL properties.
- iv. One single clockwise rotation is needed to re-establish the AVL properties.

(5 Marks)

Question 2

(a) Big-O notation provides an asymptotic measure of algorithm efficiency by focusing on the highest-order term of the time complexity function.

Consider the function:

$$T(n) = n^2 + 3n + 5$$

- i) Compute $T(n)$ for the following values of n : 10, 100, 1000, 10,000, and 100,000
- ii) Demonstrate how the contribution of the lower-order terms ($3n+5$) becomes insignificant compared to n^2 as n increases
- iii) Explain why $T(n)$ can be expressed as $O(n^2)$ in Big-O notation.

(2x3=6 Marks)

(i)

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(ii)
(iii)

b) The pseudocode below represents the recursive version of the factorial function.

```
Algorithm Factorial_Recursive(n)
  if n = 0 or n = 1 then
    return 1
  else
    return n * Factorial_Recursive(n - 1)
```

Prove that the time complexity of the above algorithm is $O(n)$

(4 Marks)

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(c) Consider the following undirected graph and algorithm, which can be used for Breadth-First Search (BFS).

BFS(u): visits all the nodes reachable from v in breadth-first order

Initialise à Queue (Q)

Mark v as visited and push it to Q

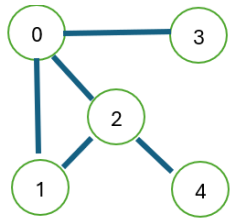
While Q is not empty:

Take the front element of Q and call it w

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For each edge $w \rightarrow u$:

If u is not visited, mark it as visited and push it to Q



Explain the Breadth-First Traversal of the above graph starting at vertex 0 using a queue.

Note: Fill in the 'Visited Nodes' and 'Queue Nodes' in the table below at the given stages (5 Marks)

Initially					
Visited					
Queue					
Step1					
Visited					
Queue					
Step 2					
Visited					
Queue					
Step 3					
Visited					
Queue					
Step4					
Visited					
Queue					
Step 4					
Visited					
Queue					

(d) Consider the following two sorted arrays:

Array A: 2, 4, 7, 11

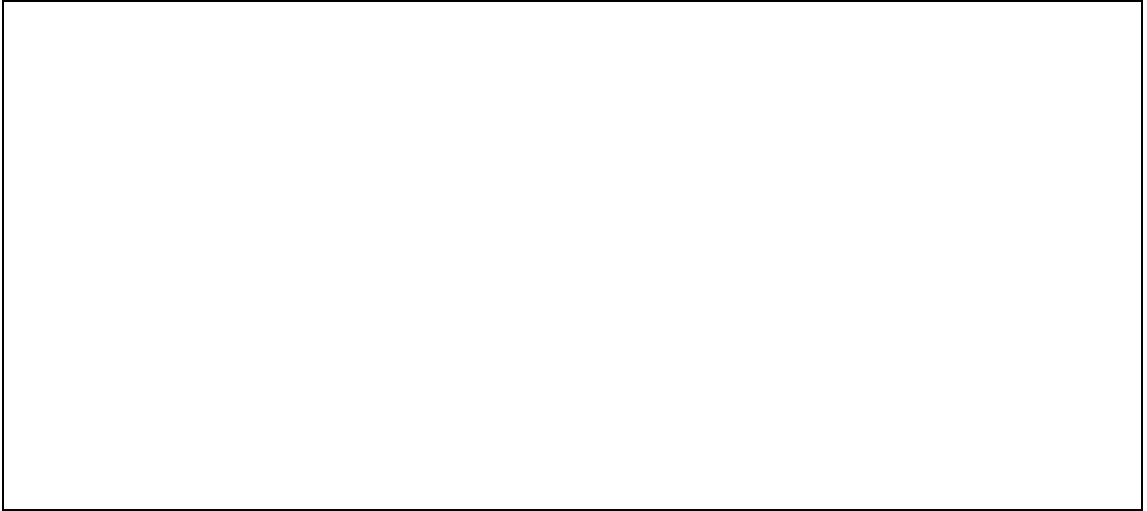
Array B: 1, 6, 8, 9, 12, 13

Describe how the above two arrays can be merged into a sorted third array, labeled as Array C.

(5 Marks)

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(e) If the above arrays (i.e., part d) are not sorted, as a preprocessing step for merging, how can you sort them using the straight merge sort approach taught during the lectures. To answer the question, you may consider the unsorted array as $D = (4, 1, 8, 5, 11, 2)$.

(5 Marks)

